Ishpreet Panesar

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EDUCATION

Toronto Metropolitan University **Expected May 2030**
Bachelor of Engineering in Mechanical Engineering

EXPERIENCE

Concrete Mix Design Member, TMU Concrete Canoe **Sep 2026 – Present**
- Assisting in the development of concrete mix designs, focusing on material composition, strength, durability, and performance under the guidance of team leads.
- Gaining familiarity with concrete batching, mixing, and specimen preparation procedures for material testing and evaluation.
- Collaborating with the mix design team to investigate concrete formulations and understand how material proportions influence mechanical properties and structural performance.

General Member, Engineers Without Borders TMU **Sept 2026 – Present**
- Participating in a student-led engineering organization focused on sustainable development, global challenges, and socially responsible engineering practices.
- Engaging with fellow engineering students to explore humanitarian engineering, interdisciplinary collaboration, and community-driven solutions.

Co-Founder, Singh Physics Tutoring **April 2025 – Present**
- Delivered personalized instruction in physics, chemistry, and math to students (grades 11–12), focusing on clear communication and individual learning needs
- Collaborated with team members to plan and execute tailored lessons, worksheets, and study materials aligned with student needs.
- Fulfilled day-to-day classroom operations, including scheduling, resource setup, and lesson delivery in a structured home-based environment.

PROJECTS

Payload Delivery RC3 Aircraft, Toronto Metropolitan Aero Design (TMAD) **Jan 2026 – Sep 2026**
- Collaborated with an 8-member engineering team to design and manufacture a scaled payload-delivery RC aircraft, integrating servos, electronic controllers, and programmable control systems.
- Developed a detailed 3D CAD model of the aircraft's tail assembly using Autodesk Fusion 360, incorporating structural and aerodynamic design considerations.
- Conducted Finite Element Analysis (FEA) on aircraft spars and landing gear to evaluate stress distribution, deformation, and structural integrity under a 6 kg static load with a safety factor of 1.5.
- Utilized XFLR5 to conduct aerodynamic simulations, analyzing airfoil characteristics, lift and drag coefficients, and aircraft stability to support design optimization.
- Manufactured and assembled the aircraft airframe using plywood, balsa wood, and Monokote covering, utilizing a custom rib-and-spar jig to maintain dimensional accuracy.
- Conducted flight testing through two successful flights carrying a 237 mL payload, evaluating aircraft performance and confirming no visible structural damage following testing.

Arthritis-Friendly Toothpaste Dispenser, *Introduction to Engineering (CEN100)* **Sep 2025 - Dec 2025**
- Collaborated with a 10-member engineering team to redesign a conventional toothpaste tube into an ergonomic pump dispenser for individuals with limited hand mobility.
- Analyzed user feedback and evaluated two pump designs based on required force, durability, performance, hygiene, and accessibility
- Contributed to a design project incorporating MATLAB-based mathematical modeling to analyze nozzle radius, fluid viscosity, yield stress, and required dispensing force.
- Supported design optimization through comparative analysis of material costs, mechanical performance, and manufacturing constraints, with the team's model identifying an optimal nozzle radius of 8.00 mm.

SKILLS

Design: Solidworks, Autodesk Fusion 360, 3D Modeling
Simulation & Analysis: Finite Element Analysis (FEA), XFLR5, Structural Analysis
Programming & Technical Tools: Python, C, MATLAB, Microsoft Excel